

Exercises for the course FM530

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The following document presents the exercises for the course FM530, taken from the book *Linear Algebra and Learning from Data* (2019) by Gilbert Strang; any page references within the exercises refer to this book.

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Exercise 1 (Section I.1 - Exercise 1). Give an example where a combination of three nonzero vectors in $\mathbb{R}^4$ is the zero vector. Then write your example in the form $Ax = 0$. What are the shapes of A and x and 0 ?

Solution 1. Let's consider the following 3 vectors:

$$\begin{aligned}a &= (1, 0, 0, 0) \\b &= (2, 0, 0, 0) \\c &= (3, 0, 0, 0)\end{aligned}$$

We clearly have that

$$a + b - c = 0$$

Let's define the matrix

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and the vector

$$x = (1, 1, 1)$$

Then we clearly have that

$$A \cdot x = (1 + 2 - 3, 0, 0, 0) = (0, 0, 0, 0)$$

Moreover, we have that $A \in \text{Mat}_{43}(\mathbb{R})$, $x \in \mathbb{R}^3$ and $0 \in \mathbb{R}^4$

Exercise 2 (Section I.1 - Exercise 4). Suppose A is the 3 by 3 matrix ones(3, 3) of all ones. Find two independent vectors x and y that solve $Ax = 0$ and $Ay = 0$. Write that first equation $Ax = 0$ (with numbers) as a combination of the columns of A . Why don't I ask for a third independent vector with $Az = 0$.

Solution 2. We have

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Using the following row-by-column operations

$$\begin{aligned} r_1 - r_2 &\rightarrow r_2 \\ r_1 - r_3 &\rightarrow r_3 \end{aligned}$$

we obtain the matrix

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and we have that $\ker(A) = \ker(B)$. Thus, to find the vectors x and y we have to find the kernel for B (that is easier). We clearly have

$$A \cdot \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0 \iff a + b + c = 0$$

Thus, fixing $c = 0$ we get $a + b = 0$ and we can chose

$$x = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

then fixing $b = 0$ we get $a + c = 0$ and we can chose

$$y = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

By construction we have $Ax = 0 = Ay$ and $\{x, y\}$ is a set of independent vectors. We also get

$$\begin{aligned} Ax = 0 &\iff \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = 0 \\ &\iff \begin{pmatrix} 1 - 1 + 0 \\ 1 - 1 + 0 \\ 1 - 1 + 0 \end{pmatrix} = 0 \\ &\iff 1 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 0 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0 \end{aligned}$$

The exercise can't ask a third independent vector z such that $Az = 0$ because we have $\ker(A) = \ker(B) = 2$.

Exercise 3 (Section I.1 - Exercise 9). Suppose the column space of an m by n matrix is all of $\mathbb{R}^3$. What can you say about m ? What can you say about n ? What can you say about the rank r ?

Solution 3. Considering the column of the matrix $A \in \text{Mat}_{mn}(\mathbb{R})$ as vectors, we get $A^1, \dots, A^n$, n vectors in $\mathbb{R}^m$. Since $\langle A^1, \dots, A^n \rangle = \mathbb{R}^3$ we have that the dimension of each vector have to be equal to 3, thus $m = 3$. Moreover, since $\dim(\mathbb{R}^3) = 3$ we have that $n \geq 3$. Thus, the matrix A is a full rank matrix.

Exercise 4 (Section I.1 - Exercise 18). If $A = CR$, what are the CR factors of the matrix $\begin{bmatrix} 0 & A \\ 0 & A \end{bmatrix}$?

Solution 4. Let's fix $A, 0 \in \text{Mat}_{nm}(\mathbb{R})$ and that $0 \in \text{Mat}_{jk}(\mathbb{R})$ (just to fix the dimension of the null matrix). It is clear that the first non zero column of

$$T = \begin{bmatrix} 0 & A \\ 0 & A \end{bmatrix}$$

is the column obtained putting the first non zero column of A above itself, thus it is the column

$$\begin{bmatrix} C^1 \\ C^1 \end{bmatrix}$$

What said above is true also for the independent columns, hence we get that the factor C for T is the matrix

$$C_T = \begin{bmatrix} C \\ C \end{bmatrix}$$

Now we have to find the factor R_T for T . We have that the first m columns of T are all 0-vectors in $\mathbb{R}^{2n}$. Hence, the first m columns of R_T have to be all 0-vector (and in order to do the row-column product the number of row for R_T have to be equal to the number of columns of C_T , that is the number of columns of C , that is the rank of A), in particular they are m 0-vector in $\mathbb{R}^{\text{rank}(A)}$. Now, as we have seen above, R_T is a matrix in $\text{Mat}_{\text{rank}(A), 2m}(\mathbb{R})$, hence we have to defining the remaining part of the matrix. Let's recall that the number of row of R is equal to the number of column of C that is the rank of A , thus it suffices to define for each $i \in \{1, \dots, m\}$, the $(m+i)$ -th column of R_T as the column

$$\begin{bmatrix} R^i \end{bmatrix}$$

Obtaining in this way that

$$R_T = \begin{bmatrix} 0_{\text{rank}(A), m} & R \end{bmatrix} \in \text{Mat}_{\text{rank}(A), 2m}(\mathbb{R})$$

where $0_{\text{rank}(A), m}$ is the null matrix in $\text{Mat}_{\text{rank}(A), m}(\mathbb{R})$. Now it's easy to see that for $j \in \{1, \dots, 2m\}$ the product between the matrix C_T and the vector R_T^j gives:

$j \leq m$: the vector $C_T \cdot 0$ (where 0 is the 0-vector in $\mathbb{R}^{\text{rank}(A)}$), that is the 0-vector in $\mathbb{R}^{\#\{\text{row of } C_T\}} = \mathbb{R}^{2\#\{\text{row of } C\}} = \mathbb{R}^{2\#\{\text{row of } A\}} = \mathbb{R}^{2n}$.

$j > m$: the vector $C_T \cdot R_T^j = C_T R^{j-m} = \begin{bmatrix} C \\ C \end{bmatrix} R^{j-m} = \begin{pmatrix} A^{j-m} \\ A^{j-m} \end{pmatrix} = T^j$.

Thus we have that $T = C_T \cdot R_T$.

Exercise 5 (Section I.2 - Exercise 2). Suppose $\mathbf{a}$ and $\mathbf{b}$ are column vectors with components $a_1, \dots, a_m$ and $b_1, \dots, b_p$. Can you multiply $\mathbf{a}$ times $\mathbf{b}^\top$ (yes or no)? What is the shape of the answer $\mathbf{a}\mathbf{b}^\top$? What number is in row i , column j of $\mathbf{a}\mathbf{b}^\top$? What can you say about $\mathbf{a}\mathbf{a}^\top$?

Solution 5. Yes, we can multiply $\mathbf{a}$ times $\mathbf{b}^\top$ and we can obtain a matrix in $\text{Mat}_{m,p}(K)$ with components:

$$(\mathbf{a} \cdot \mathbf{b}^\top)_{ij} = (a_i b_j)_{ij}$$

where $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, p\}$. Moreover we have that

$$(\mathbf{a} \cdot \mathbf{a}^\top)_{ij} = (a_i a_j)_{ij}$$

thus we have that

$$(\mathbf{a} \cdot \mathbf{a}^\top)_{ij} = (a_i a_j) = (a_j a_i) = (\mathbf{a} \cdot \mathbf{a}^\top)_{ji}$$

hence it is a symmetric matrix where the elements in the diagonal are squares (i.e. $(\mathbf{a} \cdot \mathbf{a}^\top)_{ii} = a_i^2$).

Exercise 6 (Section I.2 - Exercise 6). If A has columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ and $B = I$ is the identity matrix, what are the rank one matrices $\mathbf{a}_1 \mathbf{b}_1^*$ and $\mathbf{a}_2 \mathbf{b}_2^*$ and $\mathbf{a}_3 \mathbf{b}_3^*$? They should add to $AI = A$.

Solution 6. We clearly have that

$$\mathbf{b}_1^* = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$$

$$\mathbf{b}_2^* = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$$

$$\mathbf{b}_3^* = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

hence

$$\mathbf{a}_1 \mathbf{b}_1^* = \begin{pmatrix} a_1^1 \\ a_1^2 \\ a_1^3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} a_1^1 & 0 & 0 \\ a_1^2 & 0 & 0 \\ a_1^3 & 0 & 0 \end{pmatrix}$$

$$\mathbf{a}_2 \mathbf{b}_2^* = \begin{pmatrix} a_2^1 \\ a_2^2 \\ a_2^3 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & a_2^1 & 0 \\ 0 & a_2^2 & 0 \\ 0 & a_2^3 & 0 \end{pmatrix}$$

$$\mathbf{a}_3 \mathbf{b}_3^* = \begin{pmatrix} a_3^1 \\ a_3^2 \\ a_3^3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & a_3^1 \\ 0 & 0 & a_3^2 \\ 0 & 0 & a_3^3 \end{pmatrix}$$

and we have that

$$\mathbf{a}_1 \mathbf{b}_1^* + \mathbf{a}_2 \mathbf{b}_2^* + \mathbf{a}_3 \mathbf{b}_3^* = A$$

Exercise 7 (Section I.3 - Exercise 3). How is the nullspace of C related to the nullspaces of A and B , if $C = \begin{bmatrix} A \\ B \end{bmatrix}$?

Solution 7. Let's fix $A \in \text{Mat}_{nk}(\mathbb{R})$ and $B \in \text{Mat}_{lk}(\mathbb{R})$. Let now $x \in \mathbb{R}^k$ be a vector such that $Cx = 0$. Calling $A = (a_{ij})_{\substack{i \in \{1, \dots, n\} \\ j \in \{1, \dots, k\}}}$ and $B = (b_{ij})_{\substack{i \in \{1, \dots, l\} \\ j \in \{1, \dots, k\}}}$ we have that

$$\sum_{j=1}^k a_{i,j} x_j = 0 \quad \text{for } i \in \{1, \dots, n\}$$

and that

$$\sum_{j=1}^k b_{i,j} x_j = 0 \quad \text{for } i \in \{1, \dots, l\}$$

which means that $x \in \ker(A) \cap \ker(B)$. Conversely, holds that if $x \in \ker(A) \cap \ker(B)$, then $Cx = 0$. Then we have that $\ker(C) = \ker(A) \cap \ker(B)$.

Exercise 8 (Section I.3 - Exercise 6). (*Important*) Show that $A^\top A$ has the same nullspace as A . Here is one approach:

First, if Ax equals zero then $A^\top Ax$ equals $\mathbf{0}$. This proves $\mathbf{N}(A) \subset \mathbf{N}(A^\top A)$.

Second, if $A^\top Ax = \mathbf{0}$ then $x^\top A^\top Ax = \|Ax\|^2 = 0$. Deduce $\mathbf{N}(A^\top A) = \mathbf{N}(A)$.

Solution 8. Let's suppose that $Ax = 0$, thus it is clear that $A^\top Ax = A^\top(Ax) = A^\top 0 = 0$. Hence we have $N(A) \subset N(A^\top A)$.

Now let's suppose that $A^\top Ax = 0$. Thus we have $\|Ax\|^2 = x^\top A^\top Ax = x^\top (A^\top Ax) = x^\top 0 = 0$, since it is a norm we have $Ax = 0$. Hence $N(A^\top A) \subset N(A)$.

Exercise 9 (Section I.3 - Exercise 10). If $N(A)$ is the zero vector, what vectors are in the nullspace of $B = \begin{bmatrix} A & A & A \end{bmatrix}$?

Solution 9. Let's fix $A \in \text{Mat}_{n,m}(\mathbb{R})$. Now let's prove that $\ker(B) = \left\langle \left[\begin{array}{c} e^i \\ \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \end{array} \right]_{i \in \{1, \dots, 2m\}} \right\rangle$

where $e^i \in \mathbb{R}^{2m}$ and $\begin{pmatrix} 0 \\ \vdots \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^m$ is the vector with 1 only in the i -th entrance. Let

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in N(B)$ with $x_i \in \mathbb{R}^m$, thus $Bx = 0$, which means

$$Ax_1 + Ax_2 + Ax_3 = 0$$

Thus we have

$$0 = Ax_1 + Ax_2 + Ax_3 = A(x_1 + x_2 + x_3)$$

since $N(A) = \{0\}$ we have that $x_1 + x_2 + x_3 = 0$, that is $x_1 + x_2 = -x_3$. Thus we have the inclusion $\subset$. To prove $\supset$ suffices to observe that for any $j \in \{1, \dots, n\}$ we have

$$B_j \left[\begin{array}{c} e^i \\ \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \end{array} \right] = a_{ji} - a_{ji} = 0.$$

Exercise 10 (Section I.4 - Exercise 1). Factor these matrices into $A = LU$:

$$A = \begin{bmatrix} 2 & 1 \\ 6 & 7 \end{bmatrix} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Solution 10. We can use the following formula:

$$a_{ij} = \sum_k l_{ik} u_{kj}$$

with

$$\begin{aligned} l_{ik} &= 0 & \text{if } k > i \\ u_{kj} &= 0 & \text{if } k > j \\ l_{ii} &= 1 \end{aligned}$$

1. We get $u_{1j} = a_{1j}$ for any $j \in \{1, 2\}$, thus

$$U = \begin{pmatrix} 2 & 1 \\ 0 & * \end{pmatrix}$$

Then to compute $l_{2,1}$ we have $a_{2,1} = l_{2,1}u_{1,1} + l_{2,2}u_{2,1} \stackrel{=(u_{2,1}=0)}{=} l_{2,1}u_{1,1}$ that implies $l_{2,1} = a_{2,1}/u_{1,1} = 6/2 = 3$. Hence

$$L = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$$

Finally, to compute $u_{2,2}$ we have $a_{2,2} = l_{2,1}u_{1,2} + l_{2,2}u_{2,2}$ that implies $u_{2,2} = \frac{a_{2,2} - l_{2,1}u_{1,2}}{l_{2,2}} = \frac{1-3 \cdot 1}{1} = -2$. Thus

$$U = \begin{pmatrix} 2 & 1 \\ 0 & -2 \end{pmatrix}$$

2. as before we get

$$U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}$$

Now we can compute $l_{2,1}$ and using $a_{2,1} = l_{2,1}u_{1,1} + l_{2,2}u_{2,1} + l_{2,3}u_{3,1} \stackrel{=(u_{2,1}=0, l_{2,3}=0)}{=} l_{2,1}u_{1,1}$, that implies $l_{2,1} = a_{2,1}/u_{1,1} = 1/1 = 1$. Thus we have

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ * & * & 1 \end{pmatrix}$$

Now we can compute the second row of U ($u_{2,2}$ and $u_{2,3}$) using

$$a_{2,2} = l_{2,1}u_{1,2} + l_{2,2}u_{2,2} + l_{2,3}u_{3,2} \stackrel{=(l_{2,3}=0)}{=} l_{2,1}u_{1,2} + l_{2,2}u_{2,2}$$

and

$$a_{2,3} = l_{2,1}u_{1,3} + l_{2,2}u_{2,3} + l_{2,3}u_{3,3} \stackrel{=(l_{2,3}=0)}{=} l_{2,1}u_{1,3} + l_{2,2}u_{2,3}$$

that implies $u_{2,2} = \frac{a_{2,2}-l_{2,1}u_{1,2}}{l_{22}} = \frac{1-1 \cdot 1}{1} = 0$ and $u_{2,3} = \frac{a_{2,2}-l_{2,1}u_{1,3}}{l_{22}} = \frac{1-1 \cdot 1}{1} = 0$ getting

$$U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & * \end{pmatrix}$$

Now we can compute the $l_{3,1}$ using

$$a_{3,1} = l_{3,1}u_{1,1} + l_{3,2}u_{2,1} + l_{3,3}u_{3,1} \stackrel{(u_{2,1}=0, u_{3,1}=0)}{=} l_{3,1}u_{1,1}$$

getting $l_{3,1} = \frac{a_{3,1}}{u_{1,1}} = \frac{1}{1} = 1$. Thus we have

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & * & 1 \end{pmatrix}$$

Now we can't go further with this algorithm (indeed, the second row for U is null), but if we fix

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

and

$$U = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

we get $A = LU$ thus we have a factorization.

3. As before we get

$$U = \begin{pmatrix} 2 & -1 & 0 \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}$$

Now we compute $l_{2,1}$ using

$$a_{2,1} = l_{2,1}u_{1,1} \Rightarrow -1 = l_{2,1} \cdot 2 \Rightarrow l_{2,1} = -\frac{1}{2}$$

Thus

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ * & * & 1 \end{pmatrix}$$

Now we compute the second row of U :

$$u_{2,2} = a_{2,2} - l_{2,1}u_{1,2} = 2 - \left(-\frac{1}{2} \cdot (-1)\right) = \frac{3}{2}$$

$$u_{2,3} = a_{2,3} - l_{2,1}u_{1,3} = -1 - \left(-\frac{1}{2} \cdot 0\right) = -1$$

Thus

$$U = \begin{pmatrix} 2 & -1 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & 0 & * \end{pmatrix}$$

Now we compute $l_{3,1}$:

$$a_{3,1} = l_{3,1}u_{1,1} \Rightarrow 0 = l_{3,1} \cdot 2 \Rightarrow l_{3,1} = 0$$

Thus

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & * & 1 \end{pmatrix}$$

Now we compute $l_{3,2}$:

$$a_{3,2} = l_{3,1}u_{1,2} + l_{3,2}u_{2,2}$$

$$-1 = 0 + l_{3,2} \cdot \frac{3}{2} \Rightarrow l_{3,2} = -\frac{2}{3}$$

Thus

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{pmatrix}$$

Finally, we compute $u_{3,3}$:

$$a_{3,3} = l_{3,1}u_{1,3} + l_{3,2}u_{2,3} + u_{3,3}$$

$$2 = 0 + \left(-\frac{2}{3} \cdot -1\right) + u_{3,3}$$

$$2 = \frac{2}{3} + u_{3,3} \Rightarrow u_{3,3} = \frac{4}{3}$$

Therefore

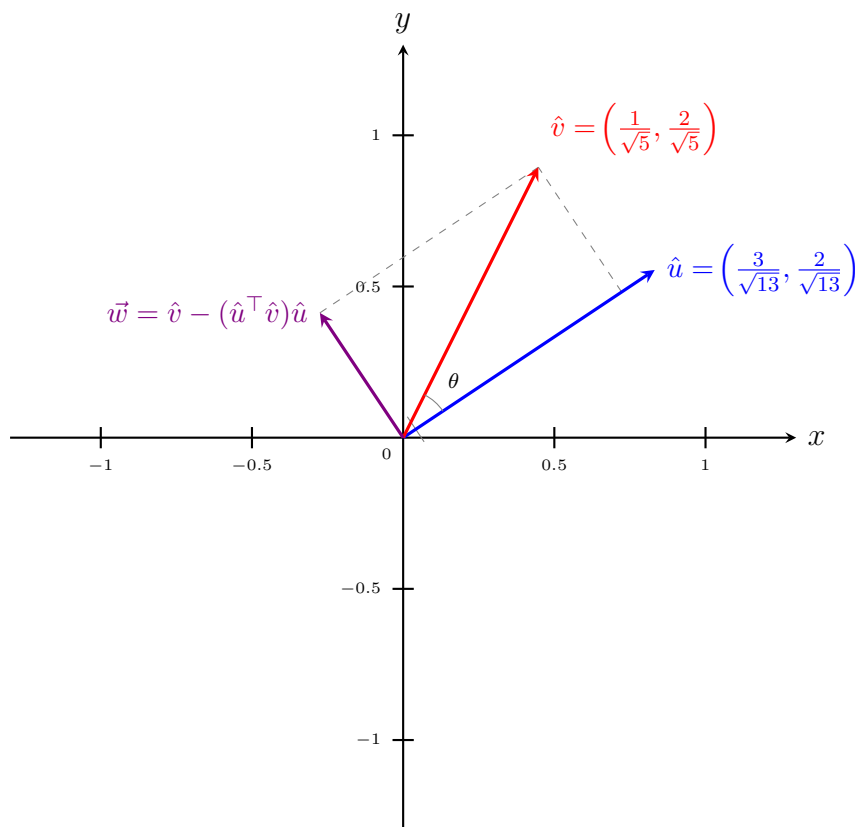
$$L = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{pmatrix} \quad U = \begin{pmatrix} 2 & -1 & 0 \\ 0 & \frac{3}{2} & -1 \\ 0 & 0 & \frac{4}{3} \end{pmatrix}$$

Exercise 11 (Section I.5 - Exercise 2). Draw unit vectors u and v that are not orthogonal. Show that $w = v - u(u^\top v)$ is orthogonal to u (and add w to your picture).

Solution 11. Let's compute $u^\top w$ and let's prove that it is zero.

$$u^\top w = u^\top (v - u(u^\top v)) = u^\top v - u^\top u(u^\top v) = (1 - u^\top u)u^\top v = (1 - \|u\|^2)u^\top v$$

Since $\|u\| = 1$ we have $\|u\|^2 = 1$ that implies $u^\top w = 0$. Then they are orthogonal.



Exercise 12 (Section I.5 - Exercise 4). Key property of every orthogonal matrix: $\|Q\mathbf{x}\|^2 = \|\mathbf{x}\|^2$ for every vector $\mathbf{x}$. More than this, show that $(Q\mathbf{x})^T(Q\mathbf{y}) = \mathbf{x}^T\mathbf{y}$ for every vector $\mathbf{x}$ and $\mathbf{y}$. So *lengths and angles are not changed by Q* . **Computations with Q never overflow!**

Solution 12. We have

$$(Qx)^T(Qy) = x^T Q^T(Qy) = x^T(Q^{-1}Q)y = x^T(\text{Id})y = x^T y$$

Exercise 13 (Section I.5 - Exercise 6). A **permutation matrix** has the same columns as the identity matrix (in some order). *Explain why this permutation matrix and every permutation matrix is orthogonal:*

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \text{ has orthonormal columns so } P^T P = \underline{\hspace{2cm}} \text{ and } P^{-1} = \underline{\hspace{2cm}}.$$

When a matrix is symmetric or orthogonal, **it will have orthogonal eigenvectors**. This is the most important source of orthogonal vectors in applied mathematics.

Solution 13. Let P a permutation matrix. We have that the rows do P^T are the columns of P , thus each elements outside of the diagonal of $P^T P$ is zero. Since $\|P^i\|^2 = 1$ we have that the elements inside the diagonal of $P^T P$ are 1's. Therefore we have

$$P^T P = \text{Id}$$

and

$$P^T = P^{-1}$$

Exercise 14 (Section I.6 - Exercise 2). Compute the eigenvalues and eigenvectors of A and A^{-1} . Check the trace!

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} -1/2 & 1 \\ 1/2 & 0 \end{bmatrix}.$$

A^{-1} has the _____ eigenvectors as A . When A has eigenvalues λ_1 and λ_2 , its inverse has eigenvalues _____.

Solution 14.

A : Let's find the roots for the following polynomial:

$$\det(A - xI_2)$$

We have that

$$(A - xI_2) = \begin{pmatrix} -x & 2 \\ 1 & 1-x \end{pmatrix}$$

that implies

$$\det(A - xI_2) = -x(1-x) - 2 = x^2 - x - 2 = (x-2)(x+1)$$

Thus the eigenvalues are

$$x_1 = 2 \quad x_2 = -1$$

Now let's find the eigenvectors using the formula

$$Az = x_i z$$

Indeed we have

$$Az = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} 2z_2 \\ z_1 + z_2 \end{pmatrix}$$

and we have to solve

$$\begin{cases} 2z_2 = 2z_1 \\ z_1 + z_2 = 2z_2 \end{cases}$$

that implies

$$z_1 = z_2$$

and the eigenvector related to the eigenvalues $x_1 = 2$ is

$$v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

To find the second eigenvector we have to solve

$$\begin{cases} 2z_2 = -z_1 \\ z_1 + z_2 = -z_2 \end{cases}$$

that implies

$$z_2 = -\frac{1}{2}z_1$$

and the eigenvector related to the eigenvalue $x_2 = -1$ is

$$v_{-1} = \begin{pmatrix} 1 \\ -\frac{1}{2} \end{pmatrix}$$

A^{-1} : Let's find the roots for the following polynomial:

$$\det(A^{-1} - xI_2)$$

We have that

$$(A^{-1} - xI_2) = \begin{pmatrix} -\frac{1}{2} - x & 1 \\ \frac{1}{2} & -x \end{pmatrix}$$

that implies

$$\det(A^{-1} - xI_2) = \left(-\frac{1}{2} - x\right)(-x) - \frac{1}{2} = \frac{x}{2} + x^2 - \frac{1}{2} = x^2 + \frac{1}{2}x - \frac{1}{2}$$

Thus the eigenvalues are

$$x_1 = \frac{1}{2} \quad x_2 = -1$$

Now let's find the eigenvectors using the formula

$$Az = x_i z$$

Indeed we have

$$A^{-1}z = \begin{pmatrix} -\frac{1}{2} & 1 \\ \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}z_1 + z_2 \\ \frac{1}{2}z_1 \end{pmatrix}$$

and we have to solve

$$\begin{cases} -\frac{1}{2}z_1 + z_2 = \frac{1}{2}z_1 \\ \frac{1}{2}z_1 = \frac{1}{2}z_2 \end{cases}$$

that implies

$$z_1 = z_2$$

and the eigenvector related to the eigenvalue $x_1 = \frac{1}{2}$ is

$$v_{\frac{1}{2}} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

To find the second eigenvector we have to solve

$$\begin{cases} -\frac{1}{2}z_1 + z_2 = -z_1 \\ \frac{1}{2}z_1 = -z_2 \end{cases}$$

that implies

$$z_1 = -2z_2$$

and the eigenvector related to the eigenvalue $x_2 = -1$ is

$$v_{-1} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

Thus the eigenvalues of the inverse matrix are the inverse of the eigenvalues of the matrix. Moreover, in both cases the trace is the sum of the eigenvalues. These two results are true for an arbitrary matrix in $\text{Mat}_2(\mathbb{R})$

Exercise 15 (Section I.6 - Exercise 11). *The eigenvalues of A equal the eigenvalues of A^T .* This is because $\det(A - \lambda I)$ equals $\det(A^T - \lambda I)$. That is true because _____. Show by an example that the eigenvectors of A and A^T are *not* the same.

Solution 15. First of all, we observe that $\det(M) = \det(M^T)$. Moreover we have that the operation $-\lambda I$ change only the diagonal of A . Hence we have that $A^T - \lambda I = (A - \lambda I)^T$. Now, let A as in the exercise 14 and let's compute the eigenvectors of A^T . We have

$$A^T z = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} z_2 \\ 2z_1 + z_2 \end{pmatrix}$$

and we have to solve

$$\begin{cases} z_2 = 2z_1 \\ 2z_1 + z_2 = 2z_2 \end{cases}$$

that implies

$$z_2 = 2z_1$$

and the eigenvector related to the eigenvalue $x_1 = 2$ is

$$v_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

To find the second eigenvector we have to solve

$$\begin{cases} z_2 = -z_1 \\ 2z_1 + z_2 = -z_2 \end{cases}$$

that implies

$$z_2 = -z_1$$

and the eigenvector related to the eigenvalue $x_2 = -1$ is

$$v_{-1} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The eigenvectors are different.

Exercise 16 (Section I.6 - Exercise 15).

(a) Factor these two matrices into $A = X\Lambda X^{-1}$:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}.$$

(b) If $A = X\Lambda X^{-1}$ then $A^3 = \begin{pmatrix} & \end{pmatrix} \begin{pmatrix} & \end{pmatrix} \begin{pmatrix} & \end{pmatrix}$ and $A^{-1} = \begin{pmatrix} & \end{pmatrix} \begin{pmatrix} & \end{pmatrix} \begin{pmatrix} & \end{pmatrix}$.

Solution 16.

(a): 1. Let's find the roots for the following polynomial:

$$\det(A - xI_2)$$

We have that

$$(A - xI_2) = \begin{pmatrix} 1-x & 2 \\ 0 & 3-x \end{pmatrix}$$

that implies

$$\det(A - xI_2) = (1-x)(3-x) = x^2 - 4x + 3 = (x-1)(x-3)$$

Thus the eigenvalues are

$$x_1 = 1 \quad x_2 = 3$$

Now let's find the eigenvectors using the formula

$$Az = x_i z$$

Indeed we have

$$Az = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} z_1 + 2z_2 \\ 3z_2 \end{pmatrix}$$

and we have to solve

$$\begin{cases} z_1 + 2z_2 = z_1 \\ 3z_2 = z_2 \end{cases}$$

that implies

$$z_2 = 0$$

and the eigenvector related to the eigenvalue $x_1 = 1$ is

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

To find the second eigenvector we have to solve

$$\begin{cases} z_1 + 2z_2 = 3z_1 \\ 3z_2 = 3z_2 \end{cases}$$

that implies

$$z_1 = z_2$$

and the eigenvector related to the eigenvalue $x_2 = 3$ is

$$v_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Then we have

$$X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$

To find the inverse of X we will use the formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

which give us

$$X^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

2. Let's find the roots for the following polynomial:

$$\det(A - xI_2)$$

We have that

$$(A - xI_2) = \begin{pmatrix} 1-x & 1 \\ 3 & 3-x \end{pmatrix}$$

that implies

$$\det(A - xI_2) = (1-x)(3-x) - 3 = x^2 - 4x + 3 - 3 = x^2 - 4x = x(x-4)$$

Thus the eigenvalues are

$$x_1 = 0 \quad x_2 = 4$$

Now let's find the eigenvectors using the formula

$$Az = x_i z$$

Indeed we have

$$Az = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \begin{pmatrix} z_1 + z_2 \\ 3z_1 + 3z_2 \end{pmatrix}$$

and we have to solve

$$\begin{cases} z_1 + z_2 = 0 \\ 3z_1 + 3z_2 = 0 \end{cases}$$

that implies

$$z_1 = -z_2$$

and the eigenvector related to the eigenvalue $x_1 = 0$ is

$$v_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

To find the second eigenvector we have to solve

$$\begin{cases} z_1 + z_2 = 4z_1 \\ 3z_1 + 3z_2 = 4z_2 \end{cases}$$

that implies

$$z_2 = 3z_1$$

and the eigenvector related to the eigenvalue $x_2 = 4$ is

$$v_4 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

Then we have

$$X = \begin{pmatrix} 1 & 1 \\ -1 & 3 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 0 & 0 \\ 0 & 3 \end{pmatrix}$$

To find the inverse of X we will use the same formula used before, which give us

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

which give us

$$X^{-1} = \frac{1}{3 - (-1)} \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} & \frac{-1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

(b): If

$$\Lambda = \begin{pmatrix} x_1 & 0 & \cdots & 0 \\ 0 & x_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & x_n \end{pmatrix}$$

we have

$$A^3 = (X\Lambda X^{-1})^3 = (X\Lambda X^{-1})(X\Lambda X^{-1})(X\Lambda X^{-1}) = X\Lambda^3 X^{-1} = X \begin{pmatrix} x_1^3 & 0 & \cdots & 0 \\ 0 & x_2^3 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & x_n^3 \end{pmatrix} X^{-1}$$

$$A^{-1} = (X\Lambda X^{-1})^{-1} = X^{-1-1} \Lambda^{-1} X^{-1} = X\Lambda^{-1} X^{-1} = X \begin{pmatrix} x_1^{-1} & 0 & \cdots & 0 \\ 0 & x_2^{-1} & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & x_n^{-1} \end{pmatrix} X^{-1}$$

Exercise 17 (Section I.7 - Exercise 3). For which numbers b and c are these matrices positive definite?

$$S = \begin{bmatrix} 1 & b \\ b & 9 \end{bmatrix} \quad S = \begin{bmatrix} 2 & 4 \\ 4 & c \end{bmatrix} \quad S = \begin{bmatrix} c & b \\ b & c \end{bmatrix}.$$

With the pivots in D and multiplier in L , factor each A into LDL^T .

Solution 17. Let's use the following test:

"All the pivots (from elimination) are positive if and only if the matrix is positive definite."

1. Let's search the pivots. We have $d_1 = 1$ and $d_2 = 9 - b^2$. Thus S is positive definite if and only if $9 - b^2 > 0$ that is true if and only if $-3 < b < 3$. Now L is built as the same as in the LU factorization, thus using the formula

$$l = \frac{s_{2,1}}{d_1}$$

we get

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 9 - b^2 \end{pmatrix} \quad L = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$$

2. Let's search the pivots. We have $d_1 = 2$ and $d_2 = c - 8$. Thus S is positive definite if and only if $c - 8 > 0$ that is true if and only if $c > 8$. Now L is built as the same as in the LU factorization, thus using the formula

$$l = \frac{s_{2,1}}{d_1}$$

we get

$$D = \begin{pmatrix} 2 & 0 \\ 0 & c - 8 \end{pmatrix} \quad L = \begin{pmatrix} 1 & 0 \\ \frac{4}{2} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

3. Let's search the pivots. We have $d_1 = c$ and $d_2 = c - \frac{b}{c}b = c - \frac{b^2}{c} = \frac{c^2 - b^2}{c}$. Thus S is positive definite if and only if $\frac{c^2 - b^2}{c} > 0$ that is true if and only if $c > 0$ (to have $d_1 > 0$) and $|b| < c$. Now L is built as the same as in the LU factorization, thus using the formula

$$l = \frac{s_{2,1}}{d_1}$$

we get

$$D = \begin{pmatrix} c & 0 \\ 0 & \frac{c^2 - b^2}{c} \end{pmatrix} \quad L = \begin{pmatrix} 1 & 0 \\ \frac{b}{c} & 1 \end{pmatrix}$$

Exercise 18 (Section I.7 - Exercise 14). Find the 3 by 3 matrix S and its pivots, rank, eigenvalues, and determinant:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} S \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 4(x_1 - x_2 + 2x_3)^2.$$

Solution 18. Let's assume that S is symmetric. First of all, let's compute the energy:

$$\begin{aligned} 4(x_1 - x_2 + 2x_3)^2 &= 4(x_1^2 + x_2^2 + 4x_3^2 - 2x_1x_2 + 4x_1x_3 - 4x_2x_3) \\ &= 4x_1^2 + 4x_2^2 + 16x_3^2 + 2(-4x_1x_2) + 2(8x_1x_3) + 2(-8x_2x_3) \end{aligned}$$

Now let's write the formula for the energy for a 3×3 matrix:

$$\begin{aligned} E(S) &= x_1^2 s_{1,1} + x_2^2 s_{2,2} + x_3^2 s_{3,3} + x_1 x_2 (s_{1,2} + s_{2,1}) + x_1 x_3 (s_{1,3} + s_{3,1}) + x_2 x_3 (s_{2,3} + s_{3,2}) \\ &= x_1^2 s_{1,1} + x_2^2 s_{2,2} + x_3^2 s_{3,3} + 2x_1 x_2 s_{1,2} + 2x_1 x_3 s_{1,3} + 2x_2 x_3 s_{2,3} \end{aligned}$$

Thus we have

$$S = \begin{pmatrix} 4 & -4 & 8 \\ -4 & 4 & -8 \\ 8 & -8 & 16 \end{pmatrix}$$

Now we can compute the determinant:

$$\begin{aligned} \det(S) &= 4(4 \cdot 16 - (-8)(-8)) - (-4)((-4) \cdot 16 - (-8) \cdot 8) + 8((-4)(-8) - 4 \cdot 8) \\ &= 4(64 - 64) - (-4)(-64 + 64) + 8(32 - 32) \\ &= 4 \cdot 0 + 4 \cdot 0 + 8 \cdot 0 = 0 \end{aligned}$$

Now we compute the pivots.

$$S_{r_2+r_1 \rightarrow r_2} = \begin{pmatrix} 4 & -4 & 8 \\ 0 & 0 & 0 \\ 8 & -8 & 16 \end{pmatrix} \quad S_{r_3-2r_1 \rightarrow r_3} = \begin{pmatrix} 4 & -4 & 8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Thus we have

$$d_1 = 4 \quad d_2 = 0 \quad d_3 = 0$$

and we have that

$$\text{rank}(S) = 1$$

To find the eigenvalues let's compute $\det(S - \text{Id}x)$.

$$\begin{aligned} \det(S - \text{Id}x) &= \det \begin{pmatrix} 4-x & -4 & 8 \\ -4 & 4-x & -8 \\ 8 & -8 & 16-x \end{pmatrix} \\ &= (4-x)[(4-x)(16-x) - (-8)(-8)] \\ &\quad - (-4)[-4(16-x) - (-8)8] \\ &\quad + 8[-4(-8) - 8(4-x)] \end{aligned}$$

Thus we have

$$\begin{aligned}\det(S - \text{Id}x) &= (4 - x) [(4 - x)(16 - x) - 64] + 4 [-4(16 - x) + 64] + 8 [32 - 8(4 - x)] \\ &= (4 - x)(x^2 - 20x) + 16x + 64x \\ &= x(4 - x)(x - 20) + 80x \\ &= x [(4 - x)(x - 20) + 80] \\ &= x [-x^2 + 24x] \\ &= -x^2(x - 24)\end{aligned}$$

we have to solve the equation

$$-x^2(x - 24) = 0$$

That implies $x_1 = 24$ and $x_2 = x_3 = 0$.

Exercise 19 (Section I.8 - Exercise 1). A symmetric matrix $S = S^T$ has orthonormal eigenvectors $\mathbf{v}_1$ to $\mathbf{v}_n$. Then any vector $\mathbf{x}$ can be written as a combination $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$. Explain these two formulas:

$$\mathbf{x}^T \mathbf{x} = c_1^2 + \cdots + c_n^2 \qquad \mathbf{x}^T S \mathbf{x} = \lambda_1 c_1^2 + \cdots + \lambda_n c_n^2.$$

Solution 19.

1. We have

$$\begin{aligned} \mathbf{x}^T \mathbf{x} &= \sum_{i=1}^n (c_1 v_1^i + \cdots + c_n v_n^i)(c_1 v_1^i + \cdots + c_n v_n^i) \\ &= \sum_{i=1}^n c_1 v_1^i (c_1 v_1^i + \cdots + c_n v_n^i) + \cdots + c_n v_n^i (c_1 v_1^i + \cdots + c_n v_n^i) \\ &= \sum_{i=1}^n c_1 v_1^i \left(\sum_{j=1}^n c_j v_j^i \right) + \cdots + c_n v_n^i \left(\sum_{j=1}^n c_j v_j^i \right) \\ &= \sum_{i=1}^n \sum_{k=1}^n c_k v_k^i \left(\sum_{j=1}^n c_j v_j^i \right) \\ &= \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n c_k v_k^i c_j v_j^i \\ &= \sum_{k=1}^n \sum_{j=1}^n \sum_{i=1}^n c_k v_k^i c_j v_j^i \\ &= \sum_{k=1}^n \sum_{j=1}^n c_k c_j \sum_{i=1}^n v_k^i v_j^i \\ &= \sum_{k=1}^n \sum_{j=1}^n c_k c_j v_k^T v_j \end{aligned}$$

Since we have that $\{v_1 \dots v_n\}$ is an orthonormal basis, we have that

$$v_k^T v_j = \begin{cases} 1, & \text{if } k = j \\ 0, & \text{if } k \neq j \end{cases}$$

we get

$$\sum_{k=1}^n \sum_{j=1}^n c_k c_j v_k^T v_j = \sum_{i=1}^n c_i^2$$

and also the claim.

2. Since S is symmetric it is also diagonalizable, thus there exists a matrix T such that

$$S = T \Lambda T^{-1}$$

where

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

Hence, using the coordinates of x in the autovector's space we get

$$x^T S x = (c_1 \quad \cdots \quad c_n) \Lambda \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}$$

and we have the claim.

Exercise 20 (Section I.8 - Exercise 6). Find the σ 's and $\mathbf{v}$'s and $\mathbf{u}$'s in the SVD for $A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix}$. Use equation (12).

Solution 20. The equation (12) (page 60 of “Gilbert Strang - Linear Algebra and Learning from Data”) is the following

$$\mathbf{U} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \quad \mathbf{\Sigma} = \begin{bmatrix} \sqrt{45} & \\ & \sqrt{5} \end{bmatrix} \quad \mathbf{V} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \quad (12)$$

and it is the decomposition in SVD for the matrix

$$A^T = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix}$$

Now we clearly have that

$$A = A^{TT} = (U\Sigma V)^T = V^T \Sigma^T U^T = V^T \Sigma U^T$$

Now, in $V_A := U^T$ we have

$$v_1 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ -3 \end{pmatrix} \quad v_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

In Σ we have $\sigma_1 = \sqrt{45}$ and $\sigma_2 = \sqrt{5}$. Finally in $U_A := V^T$ we have

$$u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Exercise 21 (Section I.9 - Exercise 2). Find a closest rank-1 approximation to these matrices (L^2 or Frobenius norm):

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad A = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} \quad A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Solution 21. From the "Eckart-Young" theorem, we have that, for any matrix B with $\text{rank}(B) = 1$ we have that

$$\|A - B\| \geq \|A - A_1\|$$

where A_1 is

$$A_1 = \sigma_1 u_1 v_1^T$$

Since we can compute A_1 , once we have done it, we get the closest rank-1 approximation.

1. Now:

σ_1 : this value is the square root of greatest eigenvalues for AA^T (or also for $A^T A$), since we have that

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

we have

$$AA^T = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and we get $\sigma_1 = \sqrt{9} = 3$.

v_1 : this vector is the eigenvector of $A^T A$ linked to σ_1 , with norm 1 and orthogonal to each v_j . Thus we have to solve

$$A^T A v_1 = \sigma_1^2 v_1$$

that is

$$(A^T A - \text{Id}_3 \sigma_1^2) v_1 = 0$$

Since A is symmetric, we get

$$A^T A = AA^T = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Thus we have

$$0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} v_1^1 \\ v_1^2 \\ v_1^3 \end{pmatrix} = \begin{pmatrix} 0 \\ v_1^2 \\ -2v_1^3 \end{pmatrix}$$

Thus we have

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

With the same computation we get $v_2 = (0, 1, 0)$ and $v_3 = (0, 0, 1)$. Thus v_1 has norm 1 and is orthogonal to the other ones.

u_1 : this vector is the eigenvector of AA^T linked to σ_1 , with norm 1 and orthogonal to each u_j . Since A is symmetric, we have that $AA^T = AA = A^T A$. Thus they have the same eigenvectors and we have

$$u_i = v_i$$

Then we have

$$A_1 = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

2. σ_1 : this value is the square root of greatest eigenvalues for AA^T (or also for $A^T A$), since we have that

$$A = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}$$

we have

$$AA^T = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$$

and we get $\sigma_1 = \sqrt{9} = 3$.

v_1 : this vector is the eigenvector of $A^T A$ linked to σ_1 , with norm 1 and orthogonal to each u_j . Thus we have to solve

$$A^T A v_1 = \sigma_1^2 v_1$$

that is

$$(A^T A - \text{Id}_2 \sigma_1^2) v_1 = 0$$

we also have

$$A^T A = \begin{bmatrix} 0 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$$

Thus we have

$$0 = \begin{bmatrix} -5 & 0 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} v_1^1 \\ v_1^2 \end{pmatrix} = \begin{pmatrix} -5v_1^1 \\ 0 \end{pmatrix}$$

Thus we have

$$v_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

With the same computation we get $v_2 = (1, 0)$. Thus v_1 has norm 1 and is orthogonal to the other one.

u_1 : this vector is the eigenvector of AA^T linked to σ_1 , with norm 1 and orthogonal to each u_j . Thus we have to solve

$$AA^T u_1 = \sigma_1^2 u_1$$

that is

$$(AA^T - \text{Id}_2 \sigma_1^2)u_1 = 0$$

Before we have computed

$$AA^T = \begin{bmatrix} 9 & 0 \\ 0 & 4 \end{bmatrix}$$

Thus we have

$$0 = \begin{bmatrix} 0 & 0 \\ 0 & -5 \end{bmatrix} \begin{pmatrix} u_1^1 \\ u_1^2 \end{pmatrix} = \begin{pmatrix} 0 \\ -5u_1^2 \end{pmatrix}$$

Thus we have

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

With the same computation we get $u_2 = (0, 1)$. Thus u_1 has norm 1 and is orthogonal to the other one.

Then we have

$$A_1 = 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} (0 \ 1) = 3 \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

3. σ_1 : this value is the square root of greatest eigenvalues for AA^T (or also for $A^T A$), since we have that

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

we have

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

Then we have

$$\det(AA^T - \text{Id}_2 x) = \begin{vmatrix} 5-x & 4 \\ 4 & 5-x \end{vmatrix} = (5-x)^2 - 16 = x^2 - 10x + 25 - 16 = x^2 - 10x + 9$$

Thus, the eigenvectors are the root of the above polynomial, that are

$$x_1 = 9 \quad x_2 = 1$$

and

$$\sigma_1 = 3$$

- v_1 : this vector is the eigenvector of $A^T A$ linked to σ_1 , with norm 1 and orthogonal to each v_j . Thus we have to solve

$$A^T A v_1 = \sigma_1^2 v_1$$

that is

$$(A^T A - \text{Id}_2 \sigma_1^2)v_1 = 0$$

Since A is symmetric, we get

$$A^T A = AA^T = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

Thus we have

$$0 = \begin{bmatrix} -4 & 4 \\ 4 & -4 \end{bmatrix} \begin{pmatrix} v_1^1 \\ v_1^2 \end{pmatrix} = \begin{pmatrix} -4v_1^1 + 4v_1^2 \\ 4v_1^1 - 4v_1^2 \end{pmatrix} 4 \begin{pmatrix} -v_1^1 + v_1^2 \\ v_1^1 - v_1^2 \end{pmatrix}$$

Thus we have

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

With the same computation we get $v_2 = (1, -1)$. Thus v_1 is orthogonal to the other ones, but we have to normalize it, then we define

$$v_1 := \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} v_1$$

u_1 : this vector is the eigenvector of AA^T linked to σ_1 , with norm 1 and orthogonal to each u_j . Since A is symmetric, we have that $AA^T = AA = A^T A$. Thus they have the same eigenvectors and we have

$$u_i = v_i$$

Then we have

$$A_1 = 3 \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} (1 \ 1) = \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Remark. One does not have to check that the vectors u_i and v_i are orthogonal to the other u_j and v_j since they are eigenvectors for the matrix $A^T A$ and AA^T , that are both symmetric, hence their eigenvectors are automatically orthogonal. Thus one has only to check the norm of the vectors.

Exercise 22 (Section I.9 - Exercise 10). If A is a 2×2 matrix with $\sigma_1 \geq \sigma_2 > 0$, find $\|A^{-1}\|_2$ and $\|A^{-1}\|_F^2$.

Solution 22. We can write A in the following way

$$A = U\Sigma V^T$$

where

$$\Sigma = \begin{pmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{pmatrix}$$

Thus we have

$$A^{-1} = V^{T^{-1}}\Sigma^{-1}U^{-1}$$

and since U and V are orthogonal matrices we get

$$A^{-1} = V\Sigma^{-1}U^T$$

Since we know that for any Q_1, Q_2 orthogonal matrix we get

$$\|Q_1 X Q_2^T\|_{2,F} = \|X\|_{2,F}$$

and since we have that U and V are orthogonal matrices (their columns are eigenvectors for a symmetric matrix) we have

$$\|A^{-1}\|_{2,F} = \|V\Sigma^{-1}U^T\|_{2,F} = \|\Sigma^{-1}\|_{2,F}$$

Now

$$\Sigma^{-1} = \begin{pmatrix} \frac{1}{\sigma_1} & 0 \\ 0 & \frac{1}{\sigma_2} \end{pmatrix}$$

then we have

$$\|\Sigma^{-1}\|_2 = \sigma_{\max}(\Sigma^{-1}) = \frac{1}{\sigma_2}$$

and

$$\|\Sigma^{-1}\|_F^2 = \sigma_1(\Sigma^{-1})^2 + \sigma_2(\Sigma^{-1})^2 = \frac{1}{\sigma_2^2} + \frac{1}{\sigma_1^2}$$

Exercise 23 (Section I.11 - Exercise 1). Show directly this fact about ℓ^1 and ℓ^2 and ℓ^∞ vector norms: $\|\mathbf{v}\|_2^2 \leq \|\mathbf{v}\|_1 \|\mathbf{v}\|_\infty$.

Solution 23. Firstly, we have

$$\|v\|_2^2 = \sum_{i=1}^n v_i^2 \quad \|v\|_1^2 = \sum_{i=1}^n |v_i| \quad \|v\|_\infty = \max_{i \in \{1, \dots, n\}} |v_i|$$

Now, since $|v_i| \leq \max_{i \in \{1, \dots, n\}} |v_i| := \|v\|_\infty$ we have the following inequality

$$v_i^2 = |v_i| |v_i| \leq \|v\|_\infty |v_i|$$

thus we get

$$\|v\|_2^2 = \sum_{i=1}^n v_i^2 \leq \sum_{i=1}^n \|v\|_\infty |v_i| = \|v\|_\infty \sum_{i=1}^n |v_i| = \|v\|_\infty \|v\|_1$$

Exercise 24 (Section I.11 - Exercise 7). A short proof of $\|AB\|_F \leq \|A\|_F \|B\|_F$ starts from multiplying rows times columns:

$$|(AB)_{ij}|^2 \leq \|\text{row } i \text{ of } A\|^2 \|\text{column } j \text{ of } B\|^2$$

is the Cauchy-Schwarz inequality.

Add up both sides over all i and j to show that $\|AB\|_F^2 \leq \|A\|_F^2 \|B\|_F^2$.

Solution 24. Let's recall that

$$\|A\|_F^2 = \sum_{i,j} a_{ij}^2$$

We clearly have

$$\begin{aligned} \|AB\|_F^2 &= \sum_{i,j} |(AB)_{ij}|^2 \\ &\leq \sum_{i,j} \|A_i\|^2 \|B^j\|^2 \\ &= \sum_{i,j} \left(\sum_{k=1}^n a_{ik}^2 \sum_{p=1}^n b_{p,j}^2 \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \left(\sum_{k=1}^n a_{ik}^2 \sum_{p=1}^n b_{p,j}^2 \right) \\ &= \left(\sum_{i=1}^n \sum_{k=1}^n a_{ik}^2 \right) \left(\sum_{j=1}^n \sum_{p=1}^n b_{p,j}^2 \right) \\ &= \|A\|_F^2 \|B\|_F^2 \end{aligned}$$

Exercise 25 (Section II.2 - Exercise 2). Why do A and A^+ have the same rank? If A is square, do A and A^+ have the same eigenvectors? What are the eigenvalues of A^+ ?

Solution 25. Let's recall that, given a matrix $A = U\Sigma V^T$ we have

$$A^+ := V\Sigma^+U^T$$

where the elements on the diagonal for Σ^+ are the $\frac{1}{\sigma_i}$.

- We have

$$\text{rank}A^+ = \text{rank}(A^{+T}) = \text{rank}(U\Sigma^+V^T) = \text{rank}(\Sigma^+)$$

Where the last equality holds since U and V are invertible matrices (matrices which columns are eigenvectors of AA^T and A^TA). Since we have

$$\frac{1}{\sigma_i} \neq 0 \iff \sigma_i \neq 0$$

and since, by the same argument, we have that

$$\text{rank}A = \text{rank}(A^T) = \text{rank}(V\Sigma U^T) = \text{rank}(\Sigma)$$

we have the claim.

- If we can consider the matrix

$$A := \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

we get

$$A^+ = \begin{pmatrix} 1/2 & 0 \\ 1/2 & 0 \end{pmatrix}$$

and they have different eigenvectors.

- If we can consider the matrix

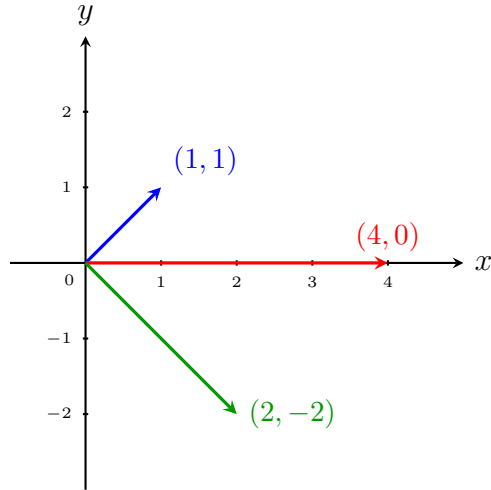
$$A := \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}$$

we get

$$A^+ = \begin{pmatrix} 2/5 & 0 \\ 1/5 & 0 \end{pmatrix}$$

On the other hand, if we consider B an invertible and symmetric matrix, we have that $B^+ = B^{-1}$. And if this holds, we have the relation that an eigenvalues λ_i for B is equal to the inverse of the eigenvalues for A^+ . Clearly, we have that the eigenvalues for the matrix A do not respect this relation, and since the relation has to be unique, can not exist a generale relation between the eigenvalues.

Exercise 26 (Section II.2 - Exercise 8). What multiple of $a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ should be subtracted from $b = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ to make the result A_2 orthogonal to a ? Sketch a figure to show a , b , and A_2 .



Solution 26. Let $k \in \mathbb{R}$ be an arbitrary element in $\mathbb{R}$. Let's consider

$$A_2 = b - ka = \begin{pmatrix} 4 \\ 0 \end{pmatrix} - \begin{pmatrix} k \\ k \end{pmatrix} = \begin{pmatrix} 4 - k \\ -k \end{pmatrix}$$

Now we impose $a^\perp A_2 = 0$ and we get the condition

$$0 = a^\perp A_2 = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 4 - k \\ -k \end{pmatrix} = 4 - k - k = 4 - 2k$$

Then we have the condition $k = 2$.

Exercise 27 (Section II.2 - Exercise 9). Complete the Gram–Schmidt process in Problem 8 by computing $q_1 = \frac{a}{\|a\|}$ and $A_2 = b - (a^\top q_1)q_1$ and $q_2 = \frac{A_2}{\|A_2\|}$ and factoring into QR :

$$\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = [q_1 \quad q_2] \begin{bmatrix} \|a\| & ? \\ 0 & \|A_2\| \end{bmatrix}.$$

Solution 27. We have

$$a = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad b = \begin{pmatrix} 4 \\ 0 \end{pmatrix}.$$

First,

$$\|a\| = \sqrt{2}, \quad q_1 = \frac{a}{\|a\|} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Now compute the projection:

$$a^\top q_1 = (1, 1) \cdot \frac{1}{\sqrt{2}}(1, 1) = \sqrt{2}.$$

Thus

$$(a^\top q_1)q_1 = \sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence

$$A_2 = b - (a^\top q_1)q_1 = \begin{pmatrix} 4 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Then

$$\|A_2\| = \sqrt{3^2 + (-1)^2} = \sqrt{10}, \quad q_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

The missing element in the matrix R is $T := q_{1,1}a_1 + q_{1,2}a_2$. Finally,

$$T = \frac{1}{\sqrt{2}}(4 + 0) = 2\sqrt{2}.$$

So the QR factorization is

$$\begin{bmatrix} 1 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{10}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{10}} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 2\sqrt{2} \\ 0 & \sqrt{10} \end{bmatrix}$$

Exercise 28 (Section II.2 - Exercise 12). With $b = 0, 8, 8, 20$ at $t = 0, 1, 3, 4$, set up and solve the normal equations $A^T A \hat{x} = A^T b$. For the best straight line in Figure II.3a, find its four heights p_i and four errors e_i . What is the minimum squared error $E = e_1^2 + e_2^2 + e_3^2 + e_4^2$?

Solution 28. By definition, we have that

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

That implies

$$A^T = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{pmatrix}$$

Thus we have

$$A^T A = \begin{pmatrix} 4 & 8 \\ 8 & 26 \end{pmatrix}$$

and

$$A^T b = \begin{pmatrix} 36 \\ 108 \end{pmatrix}$$

To solve the normal equation, we have to solve

$$\begin{pmatrix} 4 & 8 \\ 8 & 26 \end{pmatrix} \hat{x} = \begin{pmatrix} 36 \\ 108 \end{pmatrix}$$

Naming $\hat{x} = (\hat{C}, \hat{D})$ we get

$$\begin{cases} 4\hat{C} + 8\hat{D} = 36 \\ 8\hat{C} + 26\hat{D} = 108 \end{cases}$$

We get

$$\begin{aligned} & \begin{cases} \hat{C} = 9 - 2\hat{D} \\ 8\hat{C} + 26\hat{D} = 108 \end{cases} \\ & \begin{cases} \hat{C} = 9 - 2\hat{D} \\ 8(9 - 2\hat{D}) + 26\hat{D} = 108 \end{cases} \\ & \begin{cases} \hat{C} = 9 - 2\hat{D} \\ 72 + 10\hat{D} = 108 \end{cases} \\ & \begin{cases} \hat{C} = 9 - 2\hat{D} \\ \hat{D} = 36/10 \end{cases} \\ & \begin{cases} \hat{C} = 9 - 2 \cdot \frac{36}{10} = \frac{18}{10} \\ \hat{D} = \frac{36}{10} \end{cases} \end{aligned}$$

The closest line is $\frac{18}{10} + \frac{36}{10}t$ and we get

$$p_i = \frac{18}{10} + \frac{36}{10}t_i$$

and

$$\begin{cases} p_1 = \frac{18}{10} \\ p_2 = \frac{18}{10} + \frac{36}{10} = \frac{54}{10} \\ p_3 = \frac{18}{10} + 3 \cdot \frac{36}{10} = \frac{126}{10} \\ p_4 = \frac{18}{10} + 4 \cdot \frac{36}{10} = \frac{162}{10} \end{cases}$$

Moreover, since the errors are $e_i = b_i - p_i$, we get

$$\begin{cases} e_1 = -\frac{18}{10} \\ e_2 = 8 - \frac{54}{10} = \frac{80-54}{10} = \frac{26}{10} \\ e_3 = 8 - \frac{126}{10} = \frac{80-126}{10} = -\frac{46}{10} \\ e_4 = 20 - \frac{162}{10} = \frac{200-162}{10} = \frac{38}{10} \end{cases}$$

In the end, the minimum squared error is

$$\begin{aligned} E &= e_1^2 + e_2^2 + e_3^2 + e_4^2 \\ &= \left(-\frac{18}{10}\right)^2 + \left(\frac{26}{10}\right)^2 + \left(-\frac{46}{10}\right)^2 + \left(\frac{38}{10}\right)^2 \\ &= \frac{324 + 676 + 2116 + 1444}{100} \\ &= \frac{4560}{100} \\ &= \frac{228}{5} \end{aligned}$$

Exercise 29 (Section II.2 - Exercise 17). Project $\mathbf{b} = (0, 8, 8, 20)$ onto the line through $\mathbf{a} = (1, 1, 1, 1)$. Find $\hat{x} = \mathbf{a}^T \mathbf{b} / \mathbf{a}^T \mathbf{a}$ and the projection $\mathbf{p} = \hat{x} \mathbf{a}$. Check that $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is perpendicular to $\mathbf{a}$, and find the shortest distance $\|\mathbf{e}\|$ from $\mathbf{b}$ to the line through $\mathbf{a}$.

Solution 29.

$$\hat{x} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} = \frac{1 \cdot 0 + 1 \cdot 8 + 1 \cdot 8 + 1 \cdot 20}{1^2 + 1^2 + 1^2 + 1^2} = \frac{36}{4} = 9$$

$$\mathbf{p} = \hat{x} \mathbf{a} = 9 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 9 \\ 9 \\ 9 \end{bmatrix}$$

For the perpendicularity, we have

$$\mathbf{e} = \mathbf{b} - \mathbf{p} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix} - \begin{bmatrix} 9 \\ 9 \\ 9 \\ 9 \end{bmatrix} = \begin{bmatrix} -9 \\ -1 \\ -1 \\ 11 \end{bmatrix}$$

$$\mathbf{a}^T \mathbf{e} = -9 - 1 - 1 + 11 = 0$$

thus $\mathbf{e} \perp \mathbf{a}$. Moreover, for the distance we get

$$\|\mathbf{e}\| = \sqrt{81 + 1 + 1 + 121} = \sqrt{204} = 2\sqrt{51}$$

Exercise 30 (Section III.4 - Exercise 7). Find the LASSO vector x that minimizes $\frac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_1$ with

$$A = \begin{bmatrix} 4 & 1 \\ 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \lambda = 2$$

Solution 30. We have

$$\begin{aligned} Q(x) &:= \frac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_1 = \frac{1}{2}(Ax - b)^T(Ax - b) + \lambda\|x\|_1 \\ &= \frac{1}{2}(x^T A^T Ax - x^T A^T b - b^T Ax + b^T b) + \lambda(|x_1| + |x_2|) \end{aligned}$$

Then, the gradient is

$$\nabla Q(x) = A^T Ax - A^T b + \lambda \begin{pmatrix} \text{sign}(x_1) \\ \text{sign}(x_2) \end{pmatrix}$$

Let $s_1 = \text{sign}(x_1)$ and $s_2 = \text{sign}(x_2)$. Since we have

$$A^T A = \begin{bmatrix} 16 & 4 \\ 4 & 2 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

holds

$$\nabla Q(x) = \begin{bmatrix} 16 & 4 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 4 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = 0$$

Imposing $\nabla Q(x) = 0$ we get the equation

$$\begin{bmatrix} 16x_1 + 4x_2 \\ 4x_1 + 2x_2 \end{bmatrix} = \begin{bmatrix} 4 - 2s_1 \\ 2 - 2s_2 \end{bmatrix}$$

From the second equation:

$$4x_1 + 2x_2 = 2 - 2s_2 \implies x_2 = 1 - s_2 - 2x_1$$

Substituting into the first equation:

$$16x_1 + 4(1 - s_2 - 2x_1) = 4 - 2s_1$$

$$16x_1 + 4 - 4s_2 - 8x_1 = 4 - 2s_1$$

$$8x_1 = -2s_1 + 4s_2$$

$$x_1 = \frac{-s_1 + 2s_2}{4}$$

Thus we get

$$x_2 = 1 - s_2 - 2 \cdot \frac{-s_1 + 2s_2}{4} = 1 - s_2 + \frac{s_1 - 2s_2}{2} = \frac{2 - 2s_2 + s_1 - 2s_2}{2}$$

$$x_2 = \frac{s_1 - 4s_2 + 2}{2}$$

Thus, the solution is

$$\begin{cases} x_1 = \frac{-s_1 + 2s_2}{4} \\ x_2 = \frac{s_1 - 4s_2 + 2}{2} \end{cases}$$

Now let's divide in the all cases

$s_1 > 0, s_2 > 0$: We have

$$\begin{cases} x_1 = \frac{1}{4} \\ x_2 = -\frac{1}{2} \end{cases}$$

That is not possible since the signs are not compatibles.

$s_1 > 0, s_2 < 0$:

$$\begin{cases} x_1 = -\frac{3}{4} \\ x_2 = \frac{7}{2} \end{cases}$$

That is not possible since the signs are not compatibles.

$s_1 < 0, s_2 > 0$:

$$\begin{cases} x_1 = \frac{3}{4} \\ x_2 = -\frac{3}{2} \end{cases}$$

That is not possible since the signs are not compatibles.

$s_1 < 0, s_2 < 0$:

$$\begin{cases} x_1 = -\frac{1}{4} \\ x_2 = \frac{5}{2} \end{cases}$$

That is not possible since the signs are not compatibles.

This implies that at least one between x_1 and x_2 is to be zero to have a critical point. We have to study the convexity of the function in each quadrant computing the hessian matrix $H(Q)$. Since the part $\lambda(|x_1| + |x_2|)$ is linear, if we define

$$f(x) := \frac{1}{2}(x^T A^T A x - x^T A^T b - b^T A x + b^T b)$$

we get

$$H(Q(x)) = H(f(x)) = A^T A = \begin{bmatrix} 16 & 4 \\ 4 & 2 \end{bmatrix}$$

Now we have to find the eigenvalues for the matrix $A^T A$. We have

$$\det(A^T A - t \text{Id}) = \det \begin{pmatrix} 16-t & 4 \\ 4 & 2-t \end{pmatrix} = (16-t)(2-t) - 16 = t^2 - 18t + 32 - 16 = t^2 - 18t + 16$$

The eigenvalues are

$$t_{1,2} = \frac{18 \pm \sqrt{18^2 - 4 \cdot 16}}{2} = \frac{18 \pm \sqrt{324 - 64}}{2} = \frac{18 \pm \sqrt{260}}{2} = 9 \pm \sqrt{65}$$

Since $9 > \sqrt{65}$, we have that both eigenvalues are positive. Since its eigenvalues are positive, the matrix is positive definite, hence we have that $Q(x)$ is convex in each quadrant. Since we have shown that there are not critical point in any of the quadrants, by the continuity

we have that the minimum should be in the cross $x_1x_2 = 0$.

In the line $x_1 = 0$, we have to study $Q((0, x_2)) = (x_2 - 1)^2 + 2|x_2|$. We get

$$\partial_{x_2}Q((0, x_2)) = 2(x_2 - 1) + 2\text{sign}(x_2)$$

and

$$\partial_{x_2}Q((0, x_2)) = 0 \iff \begin{cases} 2x_2 - 2 + 2 = 0, & x_2 > 0 \\ 2x_2 - 2 - 2 = 0, & x_2 < 0 \end{cases} \iff \begin{cases} x_2 = 0, & x_2 > 0 \\ x_2 = 2, & x_2 < 0 \end{cases}$$

Thus we did not find any critical point.

In the line $x_2 = 0$, we have to study $Q(x_1, 0) = \frac{1}{2}((4x_1 - 1)^2 + 1) + 2|x_1|$. We get

$$\partial_{x_1}(Q(x_1, 0)) = 16x_1 - 4 + 2\text{sign}(x_1)$$

and

$$\partial_{x_1}Q((x_1, 0)) = 0 \iff \begin{cases} 16x_1 - 4 + 2 = 0, & x_1 > 0 \\ 16x_1 - 4 - 2 = 0, & x_1 < 0 \end{cases} \iff \begin{cases} x_1 = \frac{2}{16} = \frac{1}{8}, & x_1 > 0 \\ x_1 = \frac{6}{16}, & x_1 < 0 \end{cases}$$

Thus, we only have the critical point $(\frac{1}{8}, 0)$. Now, we have to compare it with the point $(0, 0)$.

$$Q\left(\frac{1}{8}, 0\right) = 8\left(\frac{1}{8}\right)^2 - 4 \cdot \frac{1}{8} + 1 + 2 \cdot \frac{1}{8} = \frac{1}{8} - \frac{1}{2} + 1 + \frac{1}{4} = \frac{1}{8} - \frac{4}{8} + \frac{8}{8} + \frac{2}{8} = \frac{7}{8}$$

$$Q(0, 0) = \frac{1}{2} \|b\|_2^2 = \frac{1}{2}(1^2 + 1^2) = 1$$

Since $Q(x^*) = \frac{7}{8} < 1 = Q(0, 0)$, the minimum is:

$$x = \begin{pmatrix} 1/8 \\ 0 \end{pmatrix}$$

Exercise 31 (Section III.4 - Exercise 8). Following Example 4, take one ADMM step in Problem 7 from $\mathbf{u}_0 = (1, 0) = \mathbf{z}_0$ with $\rho = 2$.

Solution 31. We have to use the formulas at page 188:

$$\begin{aligned} x_{k+1} &= \underset{x}{\operatorname{argmin}} \left(f(x) + \frac{1}{2}\rho \|A'x + B'z'_k - c + u_k\|^2 \right), \\ z_{k+1} &= \underset{z}{\operatorname{argmin}} \left(g(z) + \frac{1}{2}\rho \|A'x'_{k+1} + B'z - c + u_k\|^2 \right), \\ u_{k+1} &= u_k + A'x'_{k+1} + B'z'_{k+1} - c. \end{aligned}$$

Where

$$f(x) = \frac{1}{2}\|Ax - b\|^2 = \frac{1}{2}(x^T A^T Ax - x^T A^T b - b^T Ax + b^T b) = \frac{1}{2}(x^T A^T Ax - 2A^T bx + b^T b)$$

$$g(z) = \lambda \|z\|_1$$

this holds since $(2A^T bx)^T = 2A^T bx$ because its the transposed of a scalar. From the theory, we have the condition $x = z$, and we have to find A' , B' such that the equation

$$A'x + B'z = c$$

gives the previous condition. In order to do this, we have to choose $A' = \operatorname{Id}$, $B' = -\operatorname{Id}$ and $c = (0, 0)^T$. Thus we get

$$\frac{1}{2}\rho \|A'x + B'z'_0 - c + u_0\|^2 = \|x - z'_0 + u_0\|^2 = \|x\|^2$$

Thus we have

$$x_1 = \underset{x}{\operatorname{argmin}} \left(\frac{1}{2}\|Ax - b\|^2 + \|x\|^2 \right)$$

We have

$$\nabla \left(\frac{1}{2}\|Ax - b\|^2 + \|x\|^2 \right) = A^T Ax - A^T b + 2x$$

To compute the critical point we set

$$A^T Ax - A^T b + 2x = 0$$

Since we have

$$A^T Ax = \begin{pmatrix} 16x_1 + 4x_2 \\ 4x_1 + 2x_2 \end{pmatrix}, \quad A^T b = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

we have to solve the system

$$\begin{cases} 18x_1 + 4x_2 - 4 = 0 \\ 4x_1 + 4x_2 - 2 = 0 \end{cases}$$

From the second equation:

$$x_2 = \frac{2 - 4x_1}{4} = \frac{1 - 2x_1}{2}$$

Substituting in the first equation:

$$18x_1 + 4 \cdot \frac{1 - 2x_1}{2} - 4 = 0$$

$$18x_1 + 2(1 - 2x_1) - 4 = 0$$

$$18x_1 + 2 - 4x_1 - 4 = 0$$

$$14x_1 - 2 = 0$$

$$x_1 = \frac{1}{7}$$

Substituting back in the second equation:

$$x_2 = \frac{1 - 2 \cdot \frac{1}{7}}{2} = \frac{1 - \frac{2}{7}}{2} = \frac{\frac{5}{7}}{2} = \frac{5}{14}$$

Thus:

$$x = \begin{pmatrix} \frac{1}{7} \\ \frac{5}{14} \end{pmatrix}$$

Moreover, the hessian is

$$H = A^T A$$

that is a positive definite matrix, hence the matrix is convex and has as minimum the only critical point that we have found. Thus we set that

$$x'_1 = \begin{pmatrix} \frac{1}{7} \\ \frac{5}{14} \end{pmatrix}$$

We have

$$z'_1 = \underset{z}{\operatorname{argmin}} \left(g(z) + \frac{1}{2} \rho \|A'x'_1 + B'z - c + u_0\|^2 \right) = \underset{z}{\operatorname{argmin}} (\lambda \|z\|_1 + \|x'_1 - z + u_0\|^2)$$

Substituting $x'_1 = (\frac{1}{7}, \frac{5}{14})^T$ and $u_0 = (1, 0)^T$:

$$x'_1 + u_0 = \begin{pmatrix} \frac{1}{7} + 1 \\ \frac{5}{14} + 0 \end{pmatrix} = \begin{pmatrix} \frac{8}{7} \\ \frac{5}{14} \end{pmatrix}$$

We have to solve

$$z'_1 = \underset{z}{\operatorname{argmin}} (\lambda \|z\|_1 + \|x_1 - z + u_0\|^2)$$

Let $v = x_1 + u_0 = (\frac{8}{7}, \frac{5}{14})^T$. The problem separates componentwise:

$$z'_1 = \underset{z}{\operatorname{argmin}} (\lambda(|z_1| + |z_2|) + (v_1 - z_1)^2 + (v_2 - z_2)^2)$$

so we can minimize each component independently. For each component z_i we minimize

$$h(z_i) = \lambda |z_i| + (v_i - z_i)^2$$

Since h is not differentiable at $z_i = 0$, we use the subdifferential condition $0 \in \partial h(z_i)$:

$$\partial_{z_i} h(z_i) = \lambda \operatorname{sign}(z_i) - 2(v_i - z_i)$$

Component 1: We have $v_1 = \frac{8}{7}$, $\lambda = 2$.

$$2 \operatorname{sign}(z_1) - 2 \left(\frac{8}{7} - z_1 \right) = 0 \implies z_1 = \frac{8}{7} - \operatorname{sign}(z_1)$$

If $z_1 < 0$, we have that $z_1 = \frac{8}{7} - \operatorname{sign}(z_1) = \frac{15}{7}$ that is impossible. If $z_1 > 0$ we have $z_1 = \frac{8}{7} - \operatorname{sign}(z_1) = \frac{1}{7}$ that is compatible. Thus we have $z_1 = \frac{1}{7}$.

Component 2: We have $v_2 = \frac{5}{14}$, $\lambda = 2$.

$$2 \operatorname{sign}(z_2) - 2 \left(\frac{5}{14} - z_2 \right) = 0 \implies z_2 = \frac{5}{14} - \operatorname{sign}(z_2)$$

If $z_2 < 0$, we have $z_2 = \frac{5}{14} - \operatorname{sign}(z_2) = \frac{5}{14} + 1 = \frac{19}{14} > 0$, which is impossible. If $z_2 > 0$, we have $z_2 = \frac{5}{14} - \operatorname{sign}(z_2) = \frac{5}{14} - 1 = -\frac{9}{14} < 0$, which is impossible. Thus $z_2 = 0$.

We have now to compare the critical point found $z = (\frac{1}{7}, 0)$ with $(0, 0)$. Substituting both point in

$$2\|z\|_1 + \|x'_1 - z + u_0\|^2$$

we get

$$\begin{cases} z : & \frac{2}{7} + \left\| \left(\frac{1}{7}, \frac{5}{14} \right) - \left(\frac{1}{7}, 0 \right) + (1, 0) \right\|^2 = \frac{2}{7} + \left\| \left(1, \frac{5}{14} \right) \right\|^2 = \frac{2}{7} + 1 + \frac{25}{196} = \frac{2}{7} + \frac{221}{196} = \frac{56}{196} + \frac{221}{196} = \frac{277}{196} \\ (0, 0) : & 0 + \left\| \left(\frac{1}{7}, \frac{5}{14} \right) - (0, 0) + (1, 0) \right\|^2 = \left\| \left(\frac{8}{7}, \frac{5}{14} \right) \right\|^2 = \frac{64}{49} + \frac{25}{196} = \frac{256}{196} + \frac{25}{196} = \frac{281}{196} \end{cases}$$

Since $\frac{277}{196} < \frac{281}{196}$, the minimum is attained at $z_1 = (\frac{1}{7}, 0)$. Finally, we can also compute u_1 . Using the formula $u_{k+1} = u_k + A'x'_{k+1} + B'z'_{k+1} - c$ we get

$$u_1 = u_0 + x'_1 - z'_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{7} \\ \frac{5}{14} \end{pmatrix} - \begin{pmatrix} \frac{1}{7} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{5}{14} \end{pmatrix}$$

Exercise 32 (Section VI.1 - Exercise 11). Show that equation (5) is correct: Newton's Δx minimizes that quadratic.

Solution 32. We want to show that Newton's Δx_k , defined by equation (4) (page 327) as the solution of

$$H(x_k)\Delta x_k = -\nabla F(x_k)$$

minimizes the quadratic function (5)¹. Setting $\Delta x = x - x_k$, we can rewrite (5) as a function of Δx :

$$Q(\Delta x) = F(x_k) + \nabla F(x_k)^T \Delta x + \frac{1}{2} \Delta x^T H(x_k) \Delta x$$

To find the critical point we compute the gradient with respect to Δx and set it equal to zero:

$$\nabla_{\Delta x} Q(\Delta x) = \nabla F(x_k) + H(x_k) \Delta x = 0$$

which gives

$$H(x_k) \Delta x = -\nabla F(x_k)$$

Since F is convex, we get that $H(x_k)$ is positive definite, hence that it is invertible. Thus, the previous system has an only one solution that is $\Delta x = \Delta x_k$ (by definition of (4)). Moreover, since F is strictly convex, we have that $H(x_k)$ is positive definite, hence the Hessian of Q with respect to Δx is $H(x_k) > 0$, hence the critical point is a global minimum. Therefore:

$$\Delta x_k = \underset{\Delta x}{\operatorname{argmin}} \left(F(x_k) + \nabla F(x_k)^T \Delta x + \frac{1}{2} \Delta x^T H(x_k) \Delta x \right)$$

¹We recall that $H(x_k) = H(F)(x_k)$ is the hessian matrix for F evaluated in x_k .

Exercise 33 (Section VI.2 - Exercise 1). Minimize $F(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T S \mathbf{x} = \frac{1}{2}x_1^2 + 2x_2^2$ subject to $A^T \mathbf{x} = x_1 + 3x_2 = b$.

1. What is the Lagrangian $L(\mathbf{x}, \lambda)$ for this problem ?
2. What are the three equations “derivative of $L = \text{zero}$ ”?
3. Solve those equations to find $\mathbf{x}^* = (x_1^*, x_2^*)$ and the multiplier λ^* .
4. Draw Figure VI.4 for this problem with constraint line tangent to cost circle.
5. Verify that the derivative of the minimum cost is $\partial F^*/\partial b = -\lambda^*$.

Solution 33.

1. The Lagrangian for the problem is

$$L(x, \lambda) := F(x) + \lambda(a_1 x_1 + a_2 x_2 - b) = \frac{1}{2}x_1^2 + 2x_2^2 + \lambda(x_1 + 3x_2 - b)$$

2. The three equations are the followings:

$$\begin{cases} 0 = \frac{\partial L}{\partial x_1} = x_1 + \lambda \\ 0 = \frac{\partial L}{\partial x_2} = 4x_2 + 3\lambda \\ 0 = \frac{\partial L}{\partial \lambda} = x_1 + 3x_2 - b \end{cases}$$

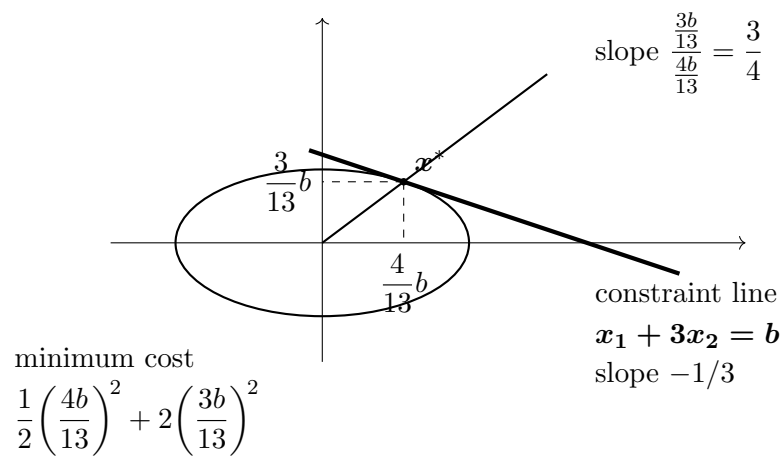
3. Now we find the solutions of the previous system

$$\begin{aligned} \begin{cases} 0 = \frac{\partial L}{\partial x_1} = x_1 + \lambda \\ 0 = \frac{\partial L}{\partial x_2} = 4x_2 + 3\lambda \\ 0 = \frac{\partial L}{\partial \lambda} = x_1 + 3x_2 - b \end{cases} &\iff \begin{cases} x_1 = -\lambda \\ x_2 = -\frac{3\lambda}{4} \\ 0 = \frac{\partial L}{\partial \lambda} = x_1 + 3x_2 - b \end{cases} \\ &\iff \begin{cases} x_1 = -\lambda \\ x_2 = -\frac{3\lambda}{4} \\ 0 = -\lambda - \frac{9}{4}\lambda - b = -\frac{13\lambda}{4} - b \end{cases} \\ &\iff \begin{cases} x_1 = -\lambda \\ x_2 = -\frac{3\lambda}{4} \\ \lambda = -\frac{4}{13}b \end{cases} \\ &\iff \begin{cases} x_1 = \frac{4}{13}b \\ x_2 = \frac{3}{13}b \\ \lambda = -\frac{4}{13}b \end{cases} \end{aligned}$$

Thus we get

$$\begin{cases} x_1^* = \frac{4}{13}b \\ x_2^* = \frac{3}{13}b \\ \lambda^* = -\frac{4}{13}b \end{cases}$$

4. The figure is



5. The minimum cost function is

$$F^*(b) = \frac{x_1^{*2}}{2} + 2x_2^{*2} = \frac{2b^2}{13}$$

The derivative is

$$\partial_b F^*(b) = \frac{4b}{13} = -\lambda^*$$

Exercise 34 (Section VI.4 - Exercise 3). Show how equations (7) and (8) lead to the basic estimate (9) for linear convergence of steepest descent. (This extends to backtracking for another choice of c .)

Solution 34. We rewrite the equations at page 350:

$$(7) : \quad f(x_{k+1}) \leq f(x_k) - \frac{1}{2M} \|\nabla f(x_k)\|^2$$

$$(8) : \quad f(x^*) \geq f(x_k) - \frac{1}{2m} \|\nabla f(x_k)\|^2$$

and recall that both M and m are strictly positive. Multiplying (7) by M we get

$$Mf(x_{k+1}) \leq Mf(x_k) - \frac{1}{2} \|\nabla f(x_k)\|^2$$

If we multiply (8) by m we get

$$mf(x^*) \geq mf(x_k) - \frac{1}{2} \|\nabla f(x_k)\|^2$$

and we change the sign we get

$$-mf(x^*) \leq -mf(x_k) + \frac{1}{2} \|\nabla f(x_k)\|^2$$

Summing these two equations, we get

$$Mf(x_{k+1}) - mf(x^*) \leq Mf(x_k) - \frac{1}{2} \|\nabla f(x_k)\|^2 - mf(x_k) + \frac{1}{2} \|\nabla f(x_k)\|^2 = Mf(x_k) - mf(x_k)$$

and we have

$$\begin{aligned} Mf(x_{k+1}) - Mf(x_k) &\leq -mf(x_k) + mf(x^*) \\ M(f(x_{k+1}) - f(x_k)) &\leq -m(f(x_k) - f(x^*)) \\ M(f(x_{k+1}) - f(x^*)) &\leq M(f(x_k) - f(x^*)) - m(f(x_k) - f(x^*)) \\ f(x_{k+1}) - f(x^*) &\leq f(x_k) - f(x^*) - \frac{m}{M}(f(x_k) - f(x^*)) \\ f(x_{k+1}) - f(x^*) &\leq \left(1 - \frac{m}{M}\right)(f(x_k) - f(x^*)) \end{aligned}$$

Exercise 35 (Section VI.4 - Exercise 6). What is the gradient descent equation $x_{k+1} = x_k - s_k \nabla f(x_k)$ for the least squares problem of minimizing $f(x) = \frac{1}{2} \|Ax - b\|^2$?

Solution 35. As we have done in the previous exercises, we have that

$$\nabla f(x) = A^T Ax - A^T b$$

thus we get the equation

$$x_{k+1} = x_k - s_k (A^T Ax - A^T b) = x_k - s_k A^T Ax_k + s_k A^T b$$

Exercise 36 (Section VI.5 - Exercise 4). Suppose we want to minimize $F(x, y) = y^2 + (y - x)^2$. The actual minimum is $F = 0$ at $(x^*, y^*) = (0, 0)$. Find the gradient vector ∇F at the starting point $(x_0, y_0) = (1, 1)$. For full gradient descent (*not stochastic*) with step $s = \frac{1}{2}$, where is (x_1, y_1) ?

Solution 36. Let's fix $x'_0 = (1, 1)$ and let's find x'_1 . The gradient is

$$\nabla F(x, y) = \begin{pmatrix} -2(y - x) \\ 2y + 2(y - x) \end{pmatrix} = \begin{pmatrix} 2x - 2y \\ 4y - 2x \end{pmatrix}$$

Thus we get

$$\nabla F(1, 1) = \begin{pmatrix} 2 - 2 \\ 4 - 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Using the formula at the beginning of page 359, that is

$$x'_{k+1} = x'_k - s \nabla F(x'_k)$$

we get

$$x'_1 = x'_0 - s \nabla F(x'_0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$